{PYTHON}

SOURCE CODE

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Complete Python Course

Learn. Build. Innovate.

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1 The Print Function

Topic 1

Introduction to Python

```
print("Hello World")
print("Welcome to Code Academy")
print("My name is Emmanuel Adedayo")
print("Today we are learning Python")
print("Starting with the first thing 'PRINT'")
```

Property of SEA Technologies Media

2 Arithmetic & Comparison Operators

Topic 2

Operators in Python

```
# ARITHMETIC OPERATORS
a = 20
b = 6

print(a + b)  #ADDITION
print(a - b)  #SUBTRACTION
print(a * b)  #MULTIPLICATION
print(a / b)  #DIVISION
print(a // b) #FLOOR DIVISION
print(a ** b) #POWER
print(a % b)  #MODULUS

print()
print()
#COMPARISON OPERATORS
print(a > b)  #True
print(a < b)  #False
print(a == b) #False
print(a != b) #True
print(a >= b) #True
print(a <= b) #False
```

3 User Input

Topic 3

Getting Input from Users

```
#INPUT
print()
print()

name = input("What is your name   ")
course = input("Which course do you like the most on this platform  ")
age = input("How old are you when you started this course  ")

print("Hello " + name + " You're welcome to this python course and am glad you love "
     + course +
      " the most, the founder will be glad to hear that. You're one of the youngest
          age " + age + " years old on this platform")
```

4 Project 1 — Simple Calculator

PROJECT Hands-On Project

Simple Calculator

A beginner project that takes two numbers from the user and performs basic arithmetic operations.

```python
#BUILDING A SIMPLE CALCULATOR
print()
print("\n---SIMPLE CALCULATOR---")
num1 = float(input("Enter first number  "))
num2 = float(input("Enter second number  "))

print("Addition:       " + int(num1 + num2))
print("Subtraction:    " + int(num1 - num2))
print("Multiplication: " + int(num1 * num2))
print("Division:       " + int(num1 / num2))
```

5 Lists & Tuples

Topic 4

Data Collections

```
#CREATING LIST
students = ["Emmanuel", "Dave", "John", "Jane", "Mike"]
courses = ["HTML", "CSS", "JavaScript", "Python", "Git", "Django"]
scores = [85, 92, 78, 65, 90]

print(students[0])
print(courses[1])
print(scores[-1])

print(len(students))

#UPDATING AN ITEM
students[0]="Emmanuel Adedayo"
print(students)

#TUPLE
days = ("Monday","Tuseday","Wednessday","Thursday","Friday","Saturday","Sunday")
print(days)
```

6 Loops

Topic 5

For Loop, While Loop & Multiplication Table

```python
# FOR LOOP
for i in range(5):
    print(i)

# for loop
for i in range (1,12):
    print("Number: " + str(i))

#for loop
courses = ("HTML", "CSS", "JAVASCRIPT", "PYTHON")
for courses in courses:
    print("I am learning " + courses)


#while loop
count = 1
while count <=100:
    print("Count is: " + str(count))
    count = count + 1

# REAL WORLD - multiplication table
print("\n--- MULTIPLICATION TABLE ---")
number = int(input("Enter a number: "))
for i in range(1, 11):
    print(str(number) + " x " + str(i) + " = " + str(number * i))
```

7 Functions

Topic 6

Defining & Calling Functions

```python
#BASIC FUNCTION
def greet():
    print("Hello, welcome to Code Academy")

greet()
greet()
greet()

#FUNCTION WITH PARAMETER
def greet_students(name):
    print("Hello " + name + " Welcome to python!")

greet_students("Emmanuel")
greet_students("Dave")
greet_students("John")


# REAL WORLD FUNCTION
def calculate_area(length, width):
    area = length * width
    return area

room1 = calculate_area(10, 5)
room2 = calculate_area(8, 6)
print("Room 1 area: " + str(room1) + " square meters")
print("Room 2 area: " + str(room2) + " square meters")
```

8 Dictionaries

Topic 7

Key-Value Pairs

```python
#CREATING A DICTIONARY
students = {
    "name": "Emmanuel Adedayo",
    "age": 20,
    "course": "python",
    "email":"seatechnologiesmedia@gmail.com",
    "is_active": True
}
print(students["name"])
print(students["age"])
print(students["course"])
print(students["email"])
print(students["is_active"])

#CHECKING IF A KEY EXIST
if "email" in students:
    print("email found " + students["email"])


# DICTIONARY OF DICTIONARIES
students = {
    "student1": {
        "name": "Emmanuel",
        "score": 85,
        "grade": "Distinction"
    },
    "student2": {
        "name": "Dave",
        "score": 72,
        "grade": "Distinction"
    },
    "student3": {
        "name": "John",
        "score": 55,
        "grade": "Pass"
    }
}

print("\n--- ALL STUDENTS ---")
for student_id, info in students.items():
    print(info["name"] + " - " + info["grade"])
```

9 Data Types

Topic 8

String, Integer, Float, Boolean

```python
#STRING
name = "Emmanuel"
course = "python"
platform = "code academy"

#INTERGER
age = 20
students = 500

#FLOAT
price = 9.99
height = 5.9
percentage = 98.5

#BOOLEAN
is_free = True
is_paid = False
is_active = True

#Print all of them
print(name)
print(course)
print(platform)
print(age)
print(students)
print(price)
print(height)
print(percentage)
print(is_free)
print(is_paid)
print(is_active)

#Check the dataype of any variable
print(type(name))
print(type(course))
print(type(platform))
print(type(age))
print(type(students))
print(type(price))
print(type(height))
print(type(percentage))
print(type(is_free))
print(type(is_paid))
print(type(is_active))
```

10 Conditional Statements

Topic 9

If, Elif, Else

```python
# BASIC IF ELSE
years = 18

if years >= 18:
    print("Congratulation You are an Adult")
else:
    print("Sorry You are not an Adult yet")


#IF ELIF ELSE
score = 75
if score >= 70:
    print("Distinction")
elif score >= 50:
    print("Pass")
elif score >= 40:
    print("Nearly Pass")
else:
    print("Fail")

# WITH USER INPUT
print("\n--- GRADE CHECKER ---")
score = float(input("Enter your score: "))

if score >= 70:
    print("Excellent! You got a Distinction")
elif score >= 50:
    print("Good job! You Passed")
elif score >= 40:
    print("Almost there! Nearly Pass")
else:
    print("Do not give up! Keep studying")

# LOGIN SYSTEM
print("\n--- LOGIN SYSTEM ---")
password = input("Enter password: ")

if password == "python123":
    print("Access granted! Welcome")
else:
    print("Wrong password. Access denied")
```

11 Project 2 — Grade Calculator

PROJECT Hands-On Project

Grade Calculator

A real-world project that calculates student grades, displays results, and saves them to a file.

```python
# ====================================================
#     GRADE CALCULATOR
#     Code Academy | SEA TECHNOLOGIES MEDIA
#     Built with Python
# ====================================================

def get_grade(average):
    if average >= 70:
        return "DISTINCTION"
    elif average >= 50:
        return "PASS"
    elif average >= 40:
        return "NEARLY PASS"
    else:
        return "FAIL"

def get_remark(average):
    if average >= 70:
        return "Excellent performance! Keep it up!"
    elif average >= 50:
        return "Good job! You can do even better!"
    elif average >= 40:
        return "Almost there! Study harder next time!"
    else:
        return "Do not give up! Keep pushing!"

def save_result(name, subjects, scores, average, grade):
    with open("results.txt", "a") as file:
        file.write("=" * 40 + "\n")
        file.write("Name: " + name + "\n")
        for i in range(len(subjects)):
            file.write(subjects[i] + ": " + str(scores[i]) + "\n")
        file.write("Average: " + str(round(average, 2)) + "\n")
        file.write("Grade: " + grade + "\n")
        file.write("=" * 40 + "\n")

def main():
    print("=" * 45)
    print("      CODE ACADEMY GRADE CALCULATOR")
    print("      SEA TECHNOLOGIES MEDIA")
    print("=" * 45)

    # Get student name
    name = input("\nEnter student full name: ")
```

```
    # Define subjects
    subjects = [
        "Mathematics",
        "English Language",
        "Python Programming",
        "Web Development",
        "Problem Solving"
    ]

    # Collect scores
    scores = []
    print("\nEnter scores for each subject (0-100):\n")

    for subject in subjects:
        while True:
            try:
                score = float(input(subject + ": "))
                if 0 <= score <= 100:
                    scores.append(score)
                    break
                else:
                    print("Score must be between 0 and 100")
            except ValueError:
                print("Please enter a valid number")

    # Calculate results
    total = sum(scores)
    average = total / len(scores)
    grade = get_grade(average)
    remark = get_remark(average)

    # Display results
    print("\n" + "=" * 45)
    print("           RESULT FOR " + name.upper())
    print("=" * 45)

    for i in range(len(subjects)):
        print(f"{subjects[i]:<25} {scores[i]}")

    print("-" * 45)
    print(f"{'Total:':<25} {total}")
    print(f"{'Average:':<25} {round(average, 2)}")
    print(f"{'Grade:':<25} {grade}")
    print("-" * 45)
    print(f"Remark: {remark}")
    print("=" * 45)

    # Save to file
    save_result(name, subjects, scores, average, grade)
    print("\nResult saved to results.txt successfully!")
    print("\nThank you for using Code Academy Grade Calculator!")
    print("Learn. Build. Innovate.")

# Run the program
main()
```